Data Cleaning & Visualization

Presented by Zaw Min Tun

Given data

ID	NAME	AGE	COUNTRY	SALARY	JOIN DATE
1	Alice	25	NZ	55,000	15/01/2020
2	Bob	NaN	NZ	60,000	20/02/2020
2	Bob	30	NZ	NaN	20/02/2020
4	Charlie	35	AUS	72,000	NaN
5	David	thirty-eight	NZ	68,000	01/11/2019
NaN	Eve	29	AU	59,000	2019-13-01
7	NaN	40	NZ	sixty five thousand	30/05/2018
8	Grace	22	NaN	64,000	25/07/2021
9	Heidi	NaN	AUS	NaN	25/07/2021
10	Ivan	27	NZ	58,000	15/03/2019

Duplicate

Null

Bad

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Cleaning

- Duplicate IDs: 1 ID(s) had multiple rows → merged (first non-null value per column retained)
- Missing ID: 1 row(s) assigned inferred sequential value(s) [3]
- Country: 2 value(s) standardised via mapping (e.g. 'AU' → 'AUS')
- Age: 1 text value(s) converted to numeric
- Salary: 1 text value(s) converted to numeric
- Join Date: 1 invalid value(s) → NaT, 1 already missing → NaT

Cleaned data

ID	NAME	AGE	COUNTRY	SALARY	JOIN DATE
1	Alice	25.0	NZ	55,000.0	2020-01-15
2	Bob	30.0	NZ	60,000.0	2020-02-20
3	Eve	29.0	AUS	59,000.0	NaT
4	Charlie	35.0	AUS	72,000.0	NaT
5	David	38.0	NZ	68,000.0	2019-11-01
7	NaN	40.0	NZ	65,000.0	2018-05-30
8	Grace	22.0	NaN	64,000.0	2021-07-25
9	Heidi	NaN	AUS	NaN	2021-07-25
10	Ivan	27.0	NZ	58,000.0	2019-03-15

Assume that merging is justified by SME

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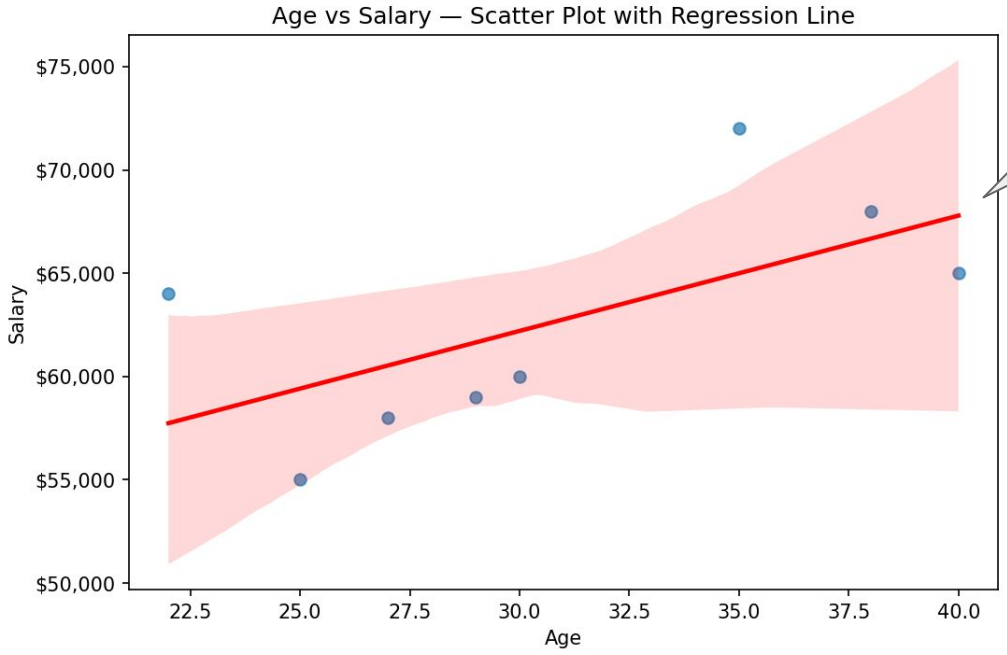
Visualization - Scatter Plot




```
Data Cleaning:
  DROPPED:
    • 2 row(s): missing Age or Salary
```

Regression line shows that with age, the salary increases.

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Visualization - Box Plot

```
Data Cleaning:
  DROPPED:
    • 1 row(s): missing Salary
    • 1 row(s): missing Country
```

AUS pays more overall.

NZ has more spread: salary varies more.

In NZ, the salaries in the lower half are more similar to each other.

No outliers detected.

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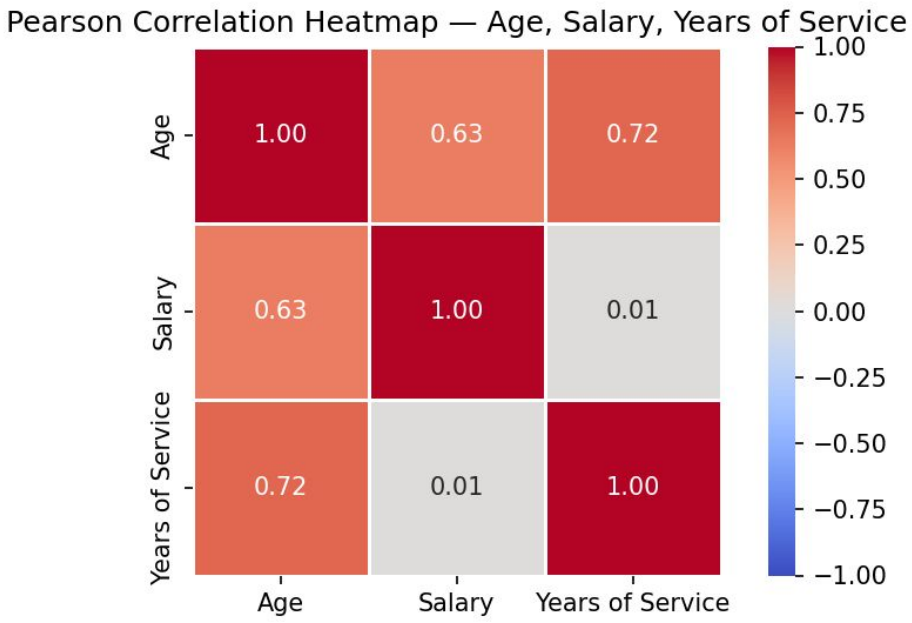
Visualization - Heatmap




```
Data Cleaning:
  DROPPED / PARTIAL:
    • 1 row(s): missing Age or Salary → excluded
    • 2 row(s): invalid/missing Join Date → 'Years of Service' is NaN (corr. uses available data only)
```

Age advances, salary increases (agreeing with the Scatter Plot)

Longer service, higher pay. (Stronger than the one above)

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Thank You!!!

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